

Week 3 - Log File Analyzer for Intrusion Detection

Abstract

Intrusion detection is a crucial component of modern cybersecurity systems. Monitoring system logs helps security teams detect suspicious activity and potential attacks before they cause significant damage. This project implements a Python-based log file analyser that scans authentication logs to identify repeated failed login attempts and detect potential brute-force attacks.

The system parses log files, extracts failed authentication attempts, counts occurrences from each IP address, and flags suspicious behavior based on predefined thresholds. A report is generated highlighting suspicious IP addresses and potential security threats. This project demonstrates how automated log analysis can help identify unauthorized access attempts and improve system security.

Introduction

System logs contain valuable information about events occurring within a computer system. These logs often include login attempts, system errors, network activities, and security-related events. Security analysts use log analysis to detect malicious activities such as brute-force attacks, unauthorized access attempts, and intrusion attempts.

Brute-force attacks occur when an attacker repeatedly attempts to guess a password by trying multiple combinations. These attacks can often be detected by analysing authentication logs for repeated failed login attempts from the same IP address.

The Log File Analyzer for Intrusion Detection is designed to simulate a basic intrusion detection system that monitors authentication logs and identifies suspicious login activity. By analysing failed login attempts, the system can detect possible brute-force attacks and generate security reports.

Tools and Technologies Used

The project was implemented using the following technologies:

Python Programming Language

Python was used to implement the log analysis logic and intrusion detection mechanism.

Regular Expressions (Regex)

Regex was used to extract IP addresses from log entries and identify failed login attempts.

Collections Module (Counter)

Python's Counter class was used to count login attempts from different IP addresses.

Text File Processing

Log files were processed using Python file handling operations.

Steps Involved in Building the Project

Step 1: Creating Sample Log Data

A sample log file was created to simulate authentication logs containing both successful and failed login attempts.

Step 2: Log Parsing

The log parser module reads the log file line by line and extracts IP addresses from failed login entries using regular expressions.

Step 3: Detecting Suspicious Activity

The detection module counts the number of failed login attempts from each IP address. If the number of attempts exceeds predefined thresholds, the IP address is marked as suspicious.

Step 4: Risk Classification

Suspicious IP addresses are classified based on severity levels such as Medium Risk or High Risk depending on the number of failed login attempts.

Step 5: Report Generation

A report generator module compiles the results and produces a readable intrusion detection report containing total attempts, IP statistics, and suspicious activity.

Conclusion

This project demonstrates how automated log analysis can be used to detect suspicious login behavior and identify potential brute-force attacks. The Log File Analyzer for Intrusion Detection provides a simple yet effective way to monitor authentication logs and highlight security threats.

Although this implementation is a simplified proof-of-concept, similar techniques are widely used in real-world Security Operations Centers (SOC) and intrusion detection systems. Future improvements could include real-time log monitoring, integration with security dashboards, and automated threat alerting.